

Lourdu Raju

Machine Learning Engineer — GenAI | Computer Vision | Production ML

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 2 years of experience building production GenAI, NLP, and Computer Vision systems at scale. Specialized in agentic AI (LangGraph, RAG, LoRA fine-tuning), low-latency model serving (vLLM, ONNX, TensorRT), and full-stack ML deployment (FastAPI, Docker, Kubernetes, AWS SageMaker). Strong CS foundations — data structures & algorithms, ML system design, and distributed systems. Kaggle Expert currently architecting Transformer-based OCR pipelines at production scale.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Bash

ML / Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, NumPy, Pandas, OpenCV, XGBoost

GenAI & LLM: LangChain, LangGraph, LlamaIndex, Ollama, vLLM; OpenAI / Anthropic / Groq APIs; LoRA fine-tuning, prompt engineering, function calling, agentic systems

RAG & Vector Search: Hybrid retrieval, dense embeddings, reranking, semantic search; ChromaDB, Qdrant, Pinecone, FAISS; sentence-transformers

Computer Vision & NLP: ViT, YOLO, CLIP; OCR (CRNN, SVRT, SVTR, PaddleOCR); BERT, RoBERTa, T5; spaCy, NLTK

MLOps, Cloud & Databases: ONNX, TensorRT, INT8 quantization; FastAPI, REST / gRPC; Docker, Kubernetes; MLflow, Weights & Biases, DVC, Git, GitHub Actions, CI / CD; AWS (EC2, Lambda, S3, SageMaker), GCP (Vertex AI); PostgreSQL, MongoDB, Redis

CS Foundations & Math: Data Structures & Algorithms, ML system design, OOP, operating systems, computer networks, DBMS; linear algebra, probability, statistics

EXPERIENCE

Sujanix Private Limited

Jan 2026 – Present

Machine Learning Engineer

Bengaluru, India

- Architected Transformer-based OCR pipelines using **SVRT** (a Swin-V2-based regression transformer) for government utility automation; improved exact-match accuracy from **82.74% to 97.04%**.
- Designed a custom **FocalCTCLoss** combining focal loss with CTC decoding to handle severe character-class imbalance; stabilized training and reduced run-to-run variance.
- Deployed via INT8 ONNX on AWS Lambda + FastAPI behind a Docker-based serving stack; reduced inference cost **30%** at **99.9%** uptime; owned the full lifecycle from labeling spec to production monitoring.

SpaceDrift

Aug 2024 – Dec 2025

Founder | MSME-registered sole proprietorship

Bengaluru, India

- Founded and operated an MSME-registered sole proprietorship delivering engineering project builds, professional data annotation services, and research support to PhD scholars on problem framing, dataset curation, and experimental pipelines.
- Spanned NLP, Computer Vision, and RAG domains as a self-funded operator; shipped **ECHOME** (agentic AI) and **FinSentinelAI** (privacy-first RAG) as flagship demonstrations.
- Engaged 3 paid contractors on a per-project basis when workload exceeded solo capacity; owned customer scoping, pricing, delivery, and stakeholder communication independently.

BrainOvision Solutions

Feb 2024 – Apr 2024

Data Science Intern

Hyderabad, India

- Improved forecasting accuracy by **15%** using ensemble gradient boosting with structured feature engineering; presented results to senior stakeholders.
- Conducted EDA on multi-million-row datasets and surfaced 3 prioritized business actions adopted by the analytics team.

PROJECTS

ECHOME — Cognitive Mirror Engine | *LangGraph, CAT/IRT, Fisher Information, XTTSv2*
github.com/Lourdhu02/echome

- Autonomous agentic AI system on LangGraph with a three-tier memory architecture (Episodic, Semantic, Procedural) for long-horizon reasoning.
- Integrated CAT / IRT psychometric models with Fisher Information maximization for adaptive assessment; reduced assessment time **70%** (30 min to 9 min).
- Privacy-first local deployment with XTTSv2 zero-shot voice cloning.

FinSentinelAI — Privacy-First Enterprise RAG | *Ollama, ChromaDB, VLMs, JWT multi-tenant*
github.com/Lourdhu02/fin-sentinal.ai

- Production-grade RAG platform with local LLMs (Ollama), ChromaDB, hybrid retrieval, faithfulness verification, and JWT-based multi-tenant session isolation.
- Multi-modal extraction from PDFs and bank statements using local Vision-Language Models for full data sovereignty.
- Built for regulated finance customers with strict on-premise / data-residency constraints.

Transformers-OCR — Industrial OCR | *SVTR, FocalCTCLoss, FRM, ONNX INT8*
github.com/Lourdhu02/transformers-ocr

- High-accuracy OCR achieving **97% exact-match** on numeric meter datasets via an SVTR backbone, custom Feature Rearrangement Modules, and FocalCTCLoss.
- ONNX INT8 quantization for edge deployment with custom Semantic Guidance Modules for robustness in motion-blur and low-contrast environments.

ACHIEVEMENTS & CERTIFICATIONS

- Kaggle Expert (Notebooks tier) — active contributor on deep learning competitions.
- Machine Learning Specialization — DeepLearning.AI / Stanford.
- NPTEL — Data Science using Python (IIT Madras).
- Microsoft Power BI Data Analyst Professional Certification.

EDUCATION

Andhra Loyola Institute of Engineering and Technology
Bachelor of Technology in Computer Science and Engineering

May 2024
Vijayawada, India